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Demystifying Urban Curbside Freight Management: Strategic Incremental Approach from Washington, D.C.

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Abstract

Loading zones and commercial vehicle activity are issues in cities across the country including the District of Columbia. Like other major U.S. cities, the District Department of Transportation (DDOT) has tried a variety of initiatives to alleviate the strain commercial vehicle activity places on the overall transportation system. These include metering or otherwise pricing loading zones and parking, expanding the size and increasing the number of loading zones, increasing the number of metered parking spaces, extending loading and metering time periods, simplifying and standardizing truck parking rules, and encouraging nighttime or off-peak goods delivery. These initiatives have had some success but a more targeted approach was needed, especially to address non-commercial vehicles parking in loading zones and thus pushing delivery vehicles to double park while loading and unloading. As part of the parkDC: Penn Quarter/Chinatown performance parking pilot, DDOT attempted to reduce the number of illegally parked cars blocking loading zones by raising prices to the highest prevailing rate on the blockface and increasing loading zone availability by extending the hours of operation of loading zones. Moving forward, DDOT has identified several next steps including: oversize/overweight vehicle routing, increased and targeted loading zone enforcement, increased disincentives for violations, data-driven modifications to the loading zone program, and modifications to the pay-by-cell program to mitigate non-trucks paying to park in loading zones. As gaps in understanding curbside movement close with increased data availability, the District expects the programs associated with curbside management to evolve to reflect smarter policy, program, and management practices.

The District of Columbia (the District or D.C.) is a constrained urban environment at the center of the sixth-largest metropolitan area in the United States. The District has a population of over 700,000, but its daytime population nearly doubles with an influx of over 500,000 commuters and visitors. Without a port or functioning freight rail yard, the District is a net consumer, not producer, of goods. Similar to many other major cities across the United States, this high concentration of activity often results in a congested transportation network, affecting personal travel and deliveries. Within the District, 99% of goods arrive by truck and businesses in the District rely heavily or solely on truck services to receive and/or ship freight.

In May 2013 the District Department of Transportation (DDOT) initiated the first District Freight Plan (*1*) to outline freight strategies and recommendations for the District to support economic growth while maintaining livability and addressing community needs and concerns. Research for the District Freight Plan found that:

- In 2011, 16.8 billion tons of domestic goods, valued at $21.7 billion, moved to, within, and from the District.
- From 2011 to 2040, freight shipments in the District are expected to grow 75% in relation to tons and 159% in relation to value.

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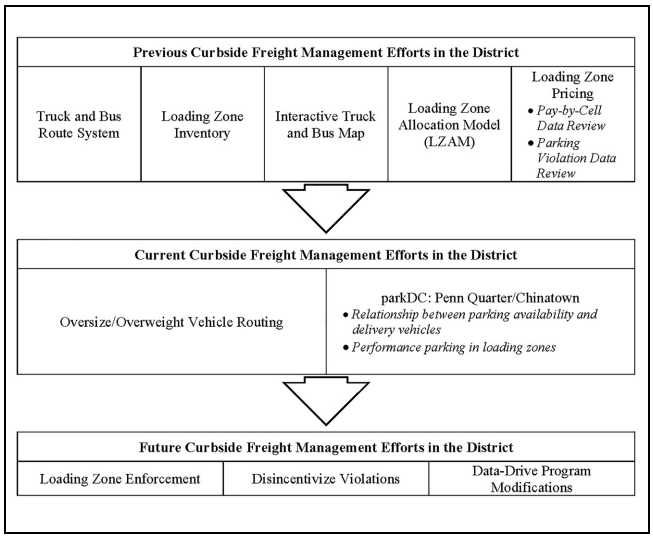


Figure 1. DDOT's strategic implementation approach to curbside freight management.

- "Inbound" and "within" freight movements are more than 25 times greater than "outbound" movements in the District.
- "Inbound" and "within" freight movements accounted for 59.3% of freight tonnage and 57.4% of the value of freight moved in, out, and through the District in 2011.

The number of trucks in the District, along with their size and operating characteristics, combined with limited available curbside loading space and a limited number of alleys all contribute to increased congestion on the District's transportation network. Both the 2003 *Mayor's Parking Taskforce Report* and the 2004 *Mayor's Downtown Congestion Task Force Final Report* identified double parking and improper loading and unloading as significant and identifiable contributors to traffic delays and included recommendations to improve overall curbside management through optimizing loading zone availability and change of current practices to enable adequate parking availability for motor carriers (*2*, *3*).

The *District Curbside Management Study*, completed in 2014, identifies the priority for management of commercial loading as ensuring access to available curbside space for goods delivery activities when it is needed, but also making this space available to other curbside uses during less active loading and delivery periods (*4*). Curbside demands and competition for curbside space has intensified as local commercial areas grow and revive. Reliably providing for commercial loading has become a greater challenge and priority, both within the District and nationally as well. Approaches contemplated by past studies or stakeholders to address this priority include:

- Metering or otherwise pricing loading zones,
- Expanding the size and increasing the number of loading zones,
- Extending loading time periods,
- Simplifying and standardizing truck parking rules, and
- Encouraging nighttime or off-peak goods delivery.

This paper describes the efforts and results of DDOT actions to implement strategically these approaches to commercial loading zones, and provides lessons learned along with next steps expected by DDOT. This strategic implementation approach is shown visually in Figure 1. Following a discussion of the literature related to urban curbside freight management, this paper follows the approach shown in Figure 1, which began in 2010 and gained momentum with the release of the *District Curbside Management Study* in 2014. Further, several additional initiatives were explored in conjunction with the parkDC: Penn Quarter/Chinatown Performance Parking pilot, which commenced in 2014 but began specifically targeting delivery vehicles in 2016. In addition to the future curbside freight management efforts, an additional ongoing study into freight trip generation will provide more detailed insights into the number and size of deliveries for typical commercial/retail uses. The results of the study will support better zoning decisions, inform future curbside management efforts, and provide a list of lessons learned to inform the state of the practice nationally.

Literature Review

Urban freight and deliveries are both a necessity and a nuisance in the urban transportation system. Issues related to the location of loading zones and pricing and time limits of loading zones have long been explored in academia. Newer loading zone management activities, such as the establishment of delivery windows and off-hour deliveries, reserving loading zones, and the use of green loading zones have been explored as well. At a high level, two practices are noted as having potential to make urban deliveries more efficient (*5*):

1. Improving on-street loading and parking facilities for trucks and service vehicles, and
2. Improving enforcement of parking regulations for personal vehicles.

Location of Loading Zones

The location of curbside loading zones is important in that it affects their use and potential illegal loading/unloading behavior. A study of optimal walking distance

from loading zones to the delivery location determined that 50 m (165 ft) is the practical maximum distance (*5*). San Francisco and New York City have designated most of the curbside in their central business districts for loading and unloading of commercial goods during business hours (*6*, *7*). Chicago allows building managers or owners to request a curbside loading zone. The Commissioner of Transportation reviews loading zone requests and, if granted, the business or manager pays an annual fee of $150 for the space which is then restricted to parking (*8*). Under the District of Columbia Municipal Regulations, businesses are required to provide a detailed loading plan including the location and use of loading berths and/or service/delivery spaces (*9*).

Pricing and Time Limits

The amount of time needed to perform a delivery or pickup can have an impact on loading availability and congestion. Pricing and time limits are often associated with each other because pricing is effectively a mechanism for enforcing time limits. Chicago does not have a time limit for commercial vehicle loading zones but does designate "15-minute standing zones" in addition to the traditional curbside loading areas (*8*). San Francisco has limited commercial vehicle loading to 30 min since the inception of the city's curbside loading zone program, and charges $ 0.75 per 30 min for the use of loading zones, though payment is not well enforced (*7*). New York City charges $2.50 for 1 h, $5 for 2 h, and $9 for 3 h of parking in most of Manhattan and Chinatown (*10*). Metered commercial vehicle parking in Midtown Manhattan is credited with reducing dwell times from 160 min to 45 min as well as reducing double parking noticeably (*11*).

Delivery Windows and Off-Hour Deliveries

New York City has established delivery windows in several neighborhoods in Brooklyn. In these areas, the New York City Department of Transportation worked with local retail merchants to identify ideal delivery windows (*10*). The municipality of São Paulo, Brazil, restricts truck access to two days a week per vehicle (*12*). Philadelphia restricted deliveries along the Walnut Street retail corridor requiring deliveries before 10:00 a.m., a delivery window the city considers a success (*11*).

Off-hour deliveries, referring to deliveries received between 7:00 p.m. and 6:00 a.m., are in effect a type of delivery window. Internationally, Barcelona, Spain, and London, U.K., have put off-hour deliveries into practice (*12*). In the United States, New York City and Los Angeles have added off-hour delivery programs to their urban freight programs (*10*, *12*), with New York City finding travel speeds improved by up to 75% in their first pilot (*10*).

Advanced Parking Management Systems

Reserving loading zones, also known as advance parking management systems, allow carriers to identify open parking spots and, in some cases, to reserve those spots. These systems resolve one of the main problems with urban deliveries—locating a spot to load and unload. When carriers can identify in advance where they can load and unload it reduces the amount of time they have to spend looking for a space or parking illegally, thus reducing congestion and emissions and facilitating more efficient deliveries (*13*). New York, Chicago, San Francisco, and Seattle have implemented a system of this type (*7*). However, this system is not popular among carriers or municipalities because of a large number of potential issues and inflexibility. When spaces are reserved, it does not allow for late deliveries, a frequent occurrence in urban freight. In addition, there is the possibility that, though the space is reserved, it is not open when the carrier arrives.

Green Loading Zones

Green loading zones (GLZs) are curbside loading zones identified for the exclusive use of electric trucks. This loading zone program, unlike the programs/strategies above, is meant to assist with curbside loading management while also encouraging the use of low-emission commercial vehicles. It is favorably viewed by municipalities as a policy because it is a low-cost option with the potential for benefits exponential to the costs (*14*). There are many options for the implementation of GLZs, including combining GLZ locations with pre-planned or existing on-street charging stations, sharing the space (allotting a desirable window of time for green vehicles and the remainder operating as a traditional loading zone), and allowing off-hour deliveries in GLZs (*14*). There also are many challenges such as utilization efficiency, the competitiveness of curbside space, and enforcement issues and challenges (*14*).

Following a European trial, a package delivery and supply chain management company has used eBikes in Pittsburgh, Pennsylvania, to deliver packages. Using electrically-assisted tricycles equipped with battery-powered electric motors, the vehicles can cover longer distances than traditional bikes, carry loads, and navigate terrain. The company places containers in central locations of the city for interim storage of packages for drivers, with final deliveries made on foot or with the eBikes (*15*).

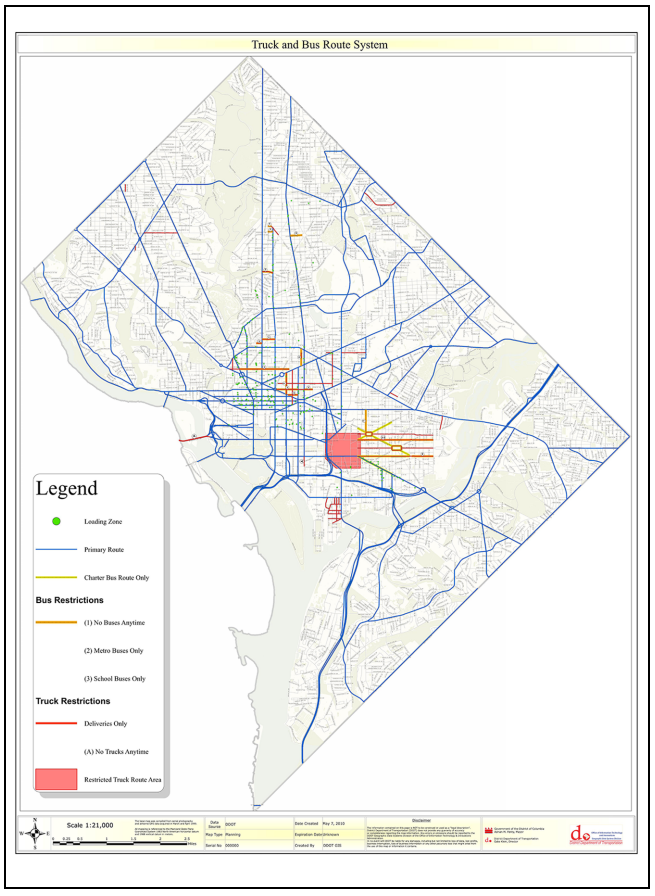


Figure 2. DDOT truck and bus route system.

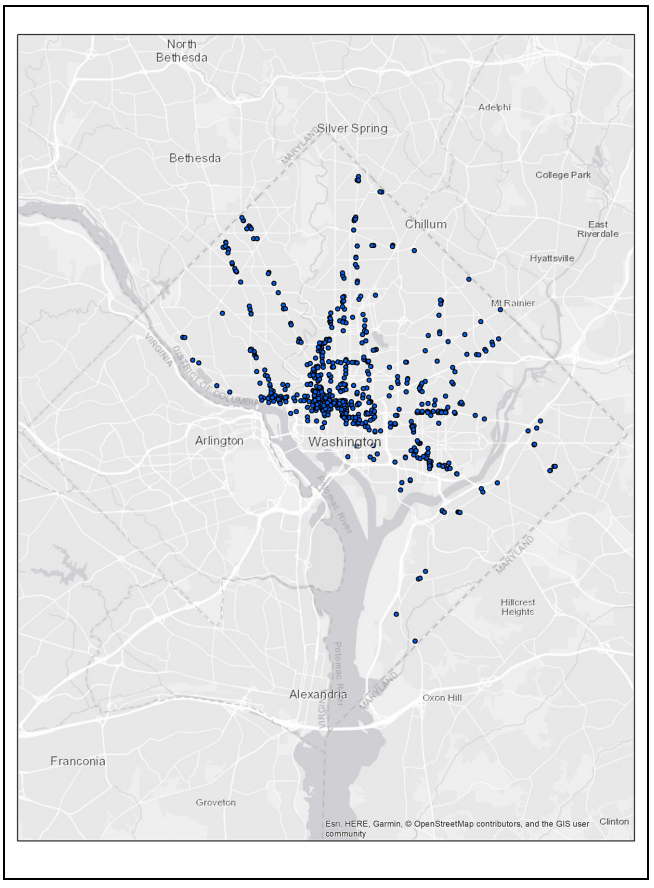


Figure 3. Loading zone inventory in Washington, D.C.

Previous Curbside Freight Management Efforts in the District

Since 2010, DDOT has moved forward with several initiatives to manage commercial vehicle activity in the District better. These efforts include the establishment of a truck and bus route system, the completion of a loading zone inventory, the development of a loading zone allocation model (LZAM), and the establishment of a program to charge for the use of all curbside commercial vehicle loading zones. These activities are described in more detail below.

Truck and Bus Route System

In 2010 the District developed a truck and bus route system to manage and monitor the truck and bus traffic in the city. Primary and restricted routes were identified based on roadway characteristics, including: roadway classification, planner review, industry and community feedback, and field observations. Primary routes are roads that can facilitate high truck traffic volumes and/or serve as major freight generators or bus destinations. Restricted routes are those that may not be used by trucks or buses because of security, inadequate capacity, or residential quality (*1*). Figure 2 displays the resulting truck and bus route system in the District.

Loading Zone Inventory

DDOT completed a comprehensive survey of commercial vehicle loading zones in the District in 2013. Using the ArcGIS Collector application, the closest address, length, side of street, curbside location, days and hours of operation, and pictures of signs and poles were collected for each of the 580 loading zones throughout the District. The loading zone inventory is displayed in Figure 3. Using this inventory, DDOT calculated that of the approximately $60.1 billion in yearly revenue generated by District businesses, $16.3 billion (27%) of business revenues are generated within 200 ft of an existing loading zone.

Interactive Truck and Bus Map

In 2013, DDOT's Motor Carrier group created a public interactive Truck and Bus Map (*16*). This map serves as a resource to the freight industry and contains truck and

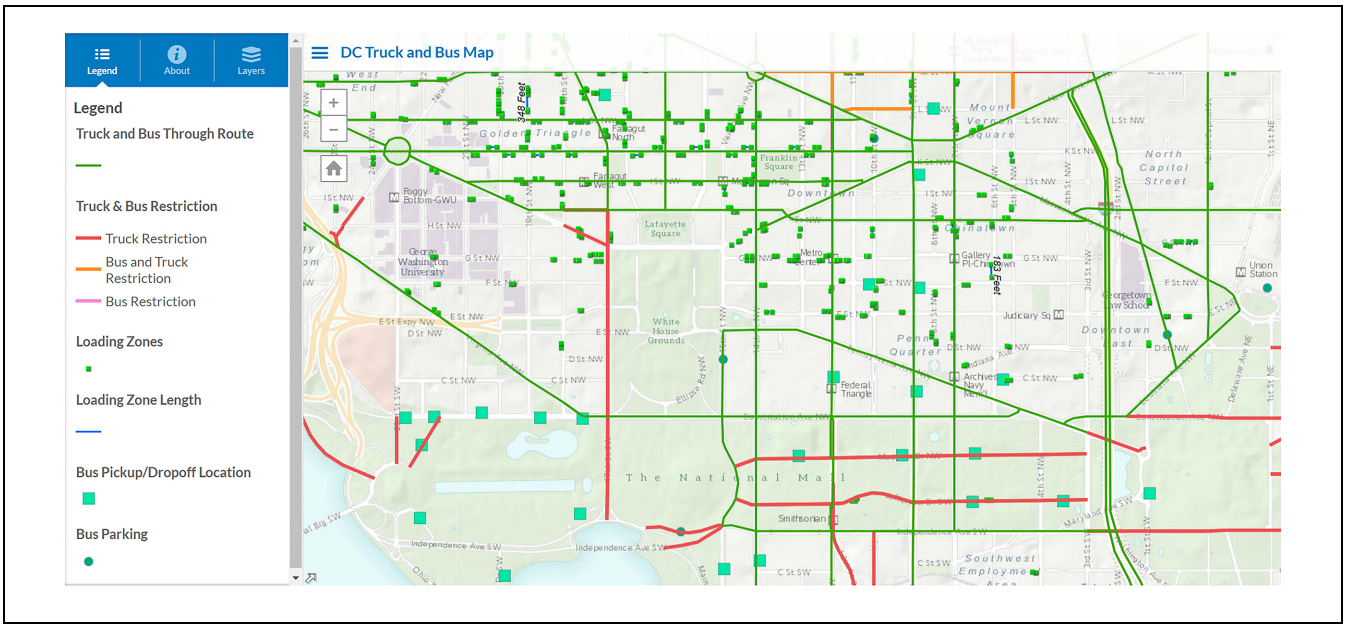


Figure 4. Interactive D.C. Truck and Bus Map (*16*).

bus routes in D.C. as well as loading zone locations and data. This map is used as an industry resource as drivers or companies can plan all aspects of their trip within the District based on the map, including routing, loading zones, and designated bus pickup and drop-off locations. A screenshot of the interactive Truck and Bus Map is displayed in Figure 4.

LZAM

In 2014, DDOT commissioned a freight trip generation model to objectively allocate and evaluate curbside loading zones. The resulting LZAM incorporates detailed urban truck-trip generation methodology using business characteristics and delivery pattern data on a block-by-block basis. The LZAM incorporates existing loading zones, gross freight trip generation (freight trip generation for the entire block face), business and building data, average building footage, and alley access. To obtain a loading zone in the District, a business or developer provides information on the expected number and size of deliveries, days and times of deliveries, and access to other potential loading space (alleys, on-site loading berths). DDOT then evaluates the request by examining curbside uses on the block and utilizing the LZAM. This provides a sustainable and repeatable process to uniformly evaluate curbside loading needs.

Loading Zone Pricing

In January 2015 the District began charging for the use of all curbside commercial vehicle loading zones. As part of the effort to charge for the use all commercial vehicle loading zones, DDOT made the decision to use an "asset-lite" approach. As part of this approach, DDOT decided against the use of loading zone specific meters and instead developed a program where carriers have the option of purchasing permits (one-day or annual) or paying for each loading zone use through a pay-by-cell (PBC) option. The permits allow commercial vehicles to occupy any loading zones in the city during their hours of operation for up to 2 h. If a loading zone is not available, the permit allows commercial vehicles to load/unload at two adjacent parking meters between the hours of 10:00 a.m. and 2:00 p.m.

As part of this initiative, DDOT updated the signage at all 580 loading zones using data from the loading zone inventory and GIS scripts that allowed for the creation of automated work orders to streamline the signage installation. Each loading zone in the District now has an assigned PBC identification number. Payment can be made by individual operators or through the PBC Fleet program (a corporate fleet program). Six months after the launch of the program 72 companies and 643 commercial vehicles registered for annual permits and 10 companies obtained 15 one-day passes. From a data collection standpoint, a drawback to this program is that DDOT does not have a mechanism for collecting data on the use of loading zones by commercial vehicles with annual or day passes.

PBC Data Review. DDOT examined PBC data to examine the types of vehicles paying to use loading zones, the

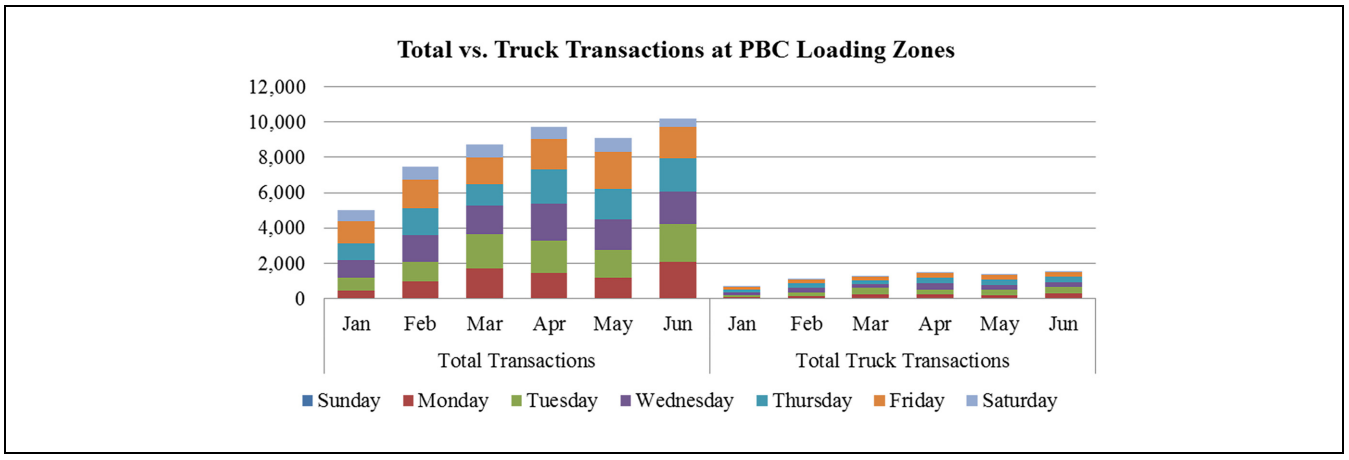


Figure 5. Total transactions versus truck transactions at PBC loading zones by month and day of week.

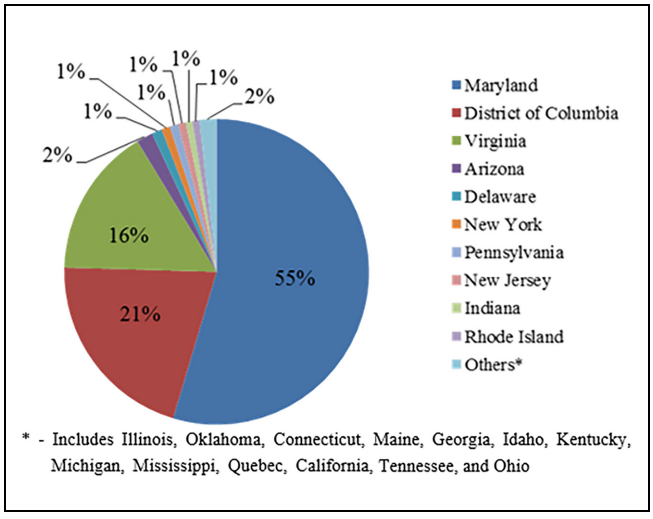


Figure 6. PBC truck user registration by jurisdiction.

states where the vehicles were registered, and the location of loading zones with the highest (and lowest) paid usage. It should be noted that the PBC data only includes vehicles paying for the use of each loading zone individually and does not include the use of loading zones by single-day or annual permit holders. PBC data includes vehicle plate information, the location of the loading zone, and the duration of paid usage (not necessarily actual usage).

DDOT reviewed six months of data (between January 1, 2015 and June 30, 2015) after DDOT began charging for the use of all curbside commercial vehicle loading zones. The data included transactions at nearly 570 loading zones and included 52,895 PBC transactions. Using license plate patterns (*17*) to identify trucks, 7,452 truck transactions (14% of all transactions in the paid loading zones) were identified. The remaining 86% of transactions at loading zones were from non-truck vehicles, including vehicles parked in loading zones legally when the loading zone is not in operation, and vehicles parked in loading zones illegally when the loading zone is in operation. The current PBC system in the District does not prohibit non-trucks from paying to park in a loading zone when the loading zone is in operation. Highlights from the PBC data review, include:

- Figure 5 shows the number of transactions by month and day of the week for all and truck subset users at loading zones during traditional loading hours. As shown, the usage of the loading zones by trucks increases as the paid loading zone program matures, however, the data also further highlights the issue related to non-trucks having the ability to pay and park in a loading zone.
- Looking at the jurisdictions where trucks using PBC are registered, as seen in Figure 6, shows that the majority of trucks using PBC in the District are registered in Maryland (55%), with 21% registered in the District, and 16% registered in Virginia.
- As shown in Figure 7, the density of both truck and non-truck loading zone activity is concentrated in the Central Business District (CBD) including the Downtown, Penn Quarter, and Chinatown neighborhoods. This is consistent with the general concentration of commercial activity in the District.
- In looking at loading zones individually, shown in Figure 8, of the 430 loading zones registering truck transactions, 17 were identified to have significant (more than two standard deviations) truck loading (more than 64 transactions in the six-month window). Of the 532 loading zones registering non-truck transactions, 35 were identified to have significant (more than two standard deviations)

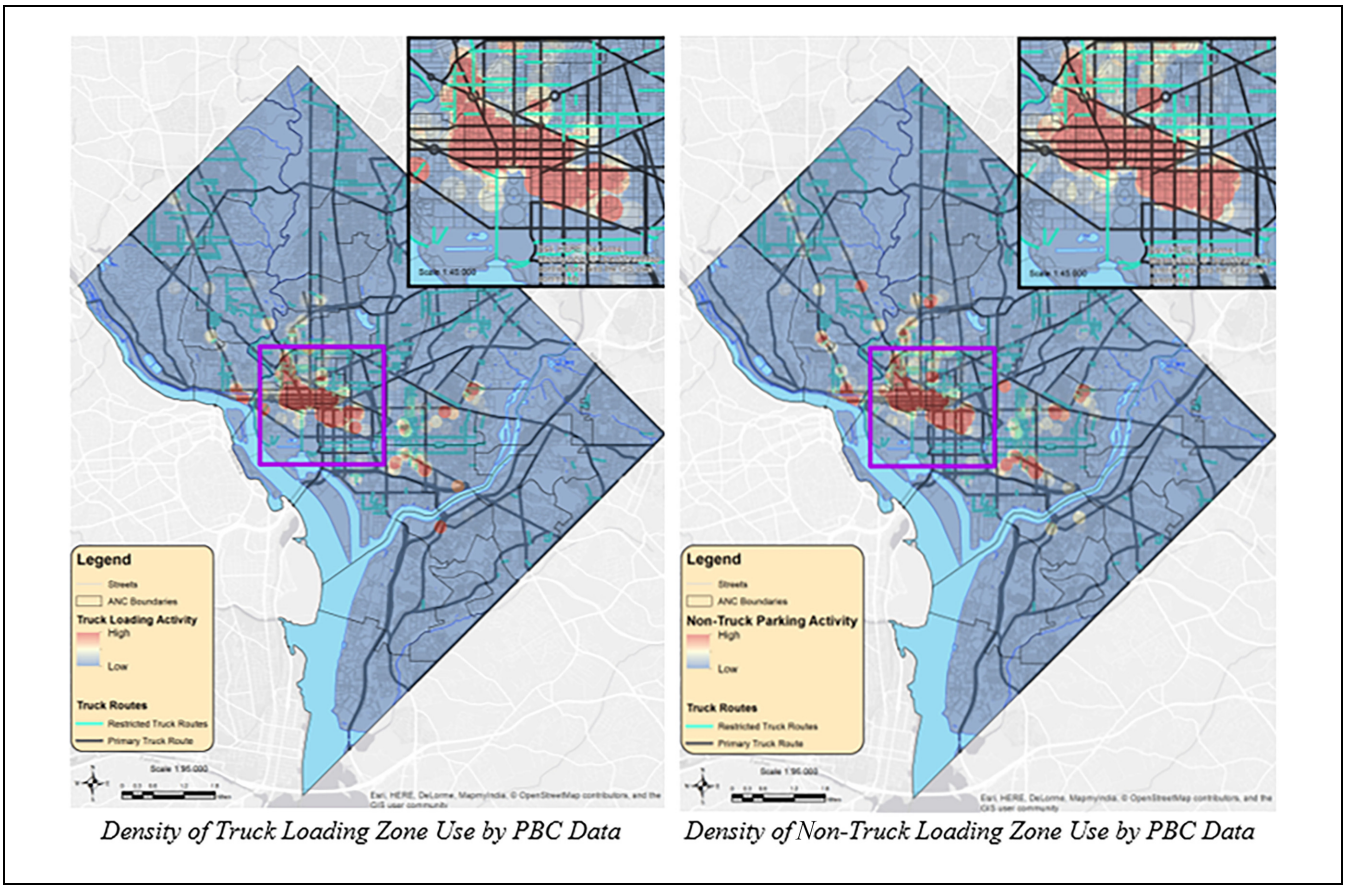


Figure 7. Density of truck and non-truck loading zone use by PBC data.

non-truck parking activity (more than 267 transactions in the six-month window).

- Among all the loading zones, nine loading zones were identified to have significant truck loading activity as well as significant non-truck parking activity. These high-usage loading zones, shown in Figure 9, are again concentrated in the CBD, and near other commercial activity centers.

Parking Violation Data Review. To complement the PBC data, DDOT also examined parking violation data provided by the District Department of Motor Vehicles, which includes the ticket number, issue date and time, violation code and description, location, and plate and state the vehicle is registered to. Data was pulled from January 1, 2009 to June 30, 2015. This data was filtered to pull only parking violations pertaining to double parking by trucks, parking in a loading zone by non-trucks, and loading zone payment and overtime violations by trucks. From nearly 9 million parking citations issued in the 6.5-year analysis period, 372,224 citations were located in reference to loading zones and truck loading violations. Until January 2015 the fine for all infractions in the dataset was $50. In January 2015, the fine for unauthorized vehicles in a loading zone was raised to $100 while truck-related fines remained at $50. Highlights from the parking violation data review include:

- Similar to the PBC data, approximately 90% of citations were issued to vehicles plated in the District of Columbia, Maryland, or Virginia.
- As shown in Figure 10, the number of violations related to loading zones decreased from 2013 to 2015 compared with the number of violations between 2009 and 2012. This is likely because of changes in enforcement actions rather than any specific program or initiative pushed by DDOT.
- Except for year 2009, double parking is the most common parking violation for trucks. Speculatively, this could be attributed to the lack of availability of loading zones because of high

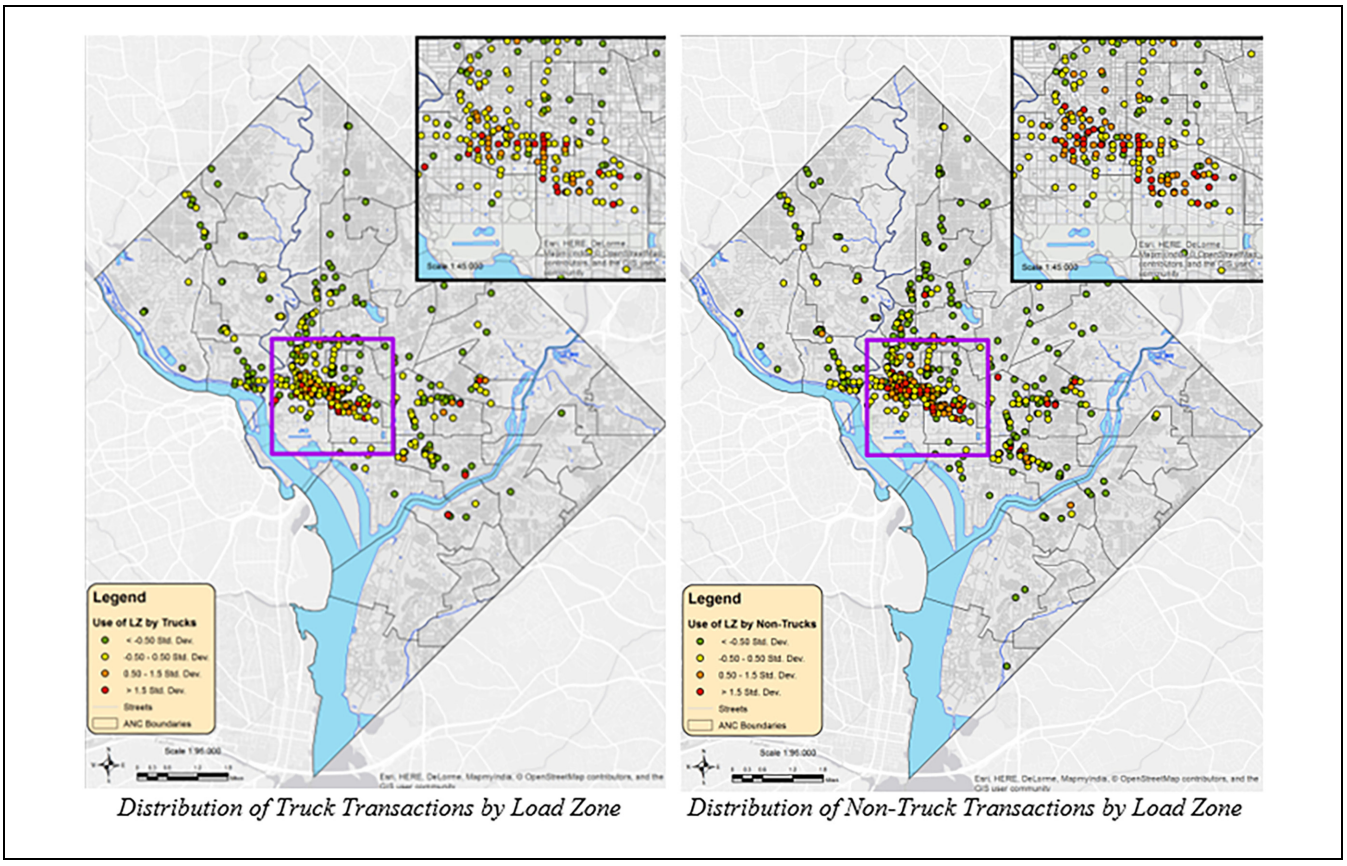


Figure 8. Distribution of transactions by trucks and non-trucks at PBC loading zones.

PBC transaction density of non-trucks in loading zones.

- When reviewing the distribution of loading zone violations involving truck double parking, non-truck parked in loading zone, and truck non-payment, it was found that the distribution of these violations correlates with the density of commercial activity centers in the District.

Loading Zone Pricing Lessons Learned. DDOT's move to pricing of all loading zones and subsequent data analysis provided lessons learned that were carried forward to current curbside freight management efforts in the District. The PBC data review confirmed industry and District knowledge, such as the originating state and destination areas, (55% of commercial vehicles paying for loading zone time are registered in Maryland) and the days and times loading most often occurs. The data also confirms that loading zones are often occupied by unauthorized vehicles and identifies where that occurs most frequently. The parking violation review made clear to DDOT that most of tickets that were being issued were to commercial vehicles for double parking, rather than to unauthorized vehicles. This information informed changes in enforcement strategy including additional coordination with enforcement agencies. The use of loading zones by unauthorized vehicles seen in the PBC data and the resulting double parking by commercial vehicles seen in the parking violation data both helped guide the freight management efforts explored as part of the parkDC: Penn Quarter/Chinatown project discussed in the next section.

Current Curbside Freight Management Efforts in the District

Oversize/Overweight Vehicle Routing

Building on the public interactive Truck and Bus Map (Figure 4), the District has developed an enterprise routing tool (*18*) where the operators of oversize and overweight (OSOW) vehicles enter their truck load and dimension details into a tool embedded in the permitting process and are routed through and out of the District on streets that can handle the vehicle and weight for that trip. This tool is available publicly; drivers and companies not required to obtain an OSOW vehicle permit are able to utilize it for routing purposes as well. A

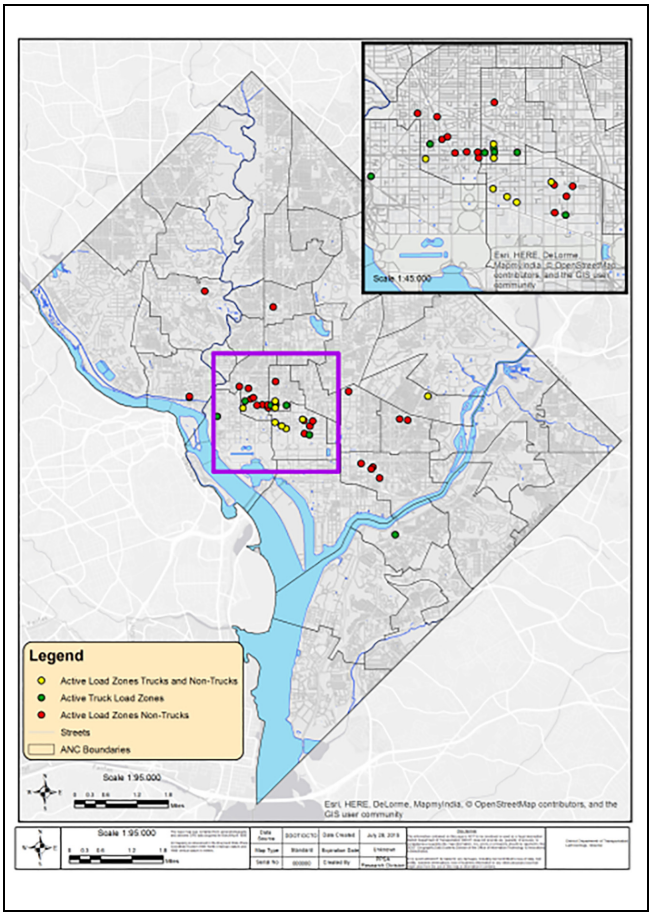


Figure 9. High use (more than two standard deviations), truck and non-truck, by loading zone.

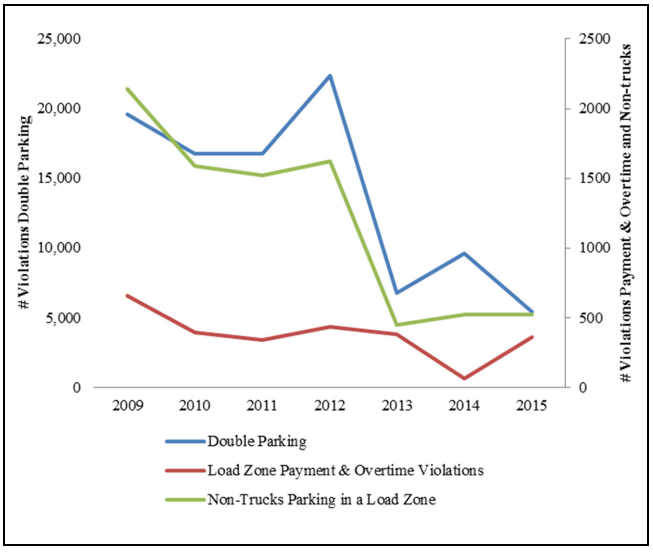


Figure 10. Aggregate annual trends of parking violations, 2009 to 2015.

screenshot of the interactive routing tool is displayed in Figure 11.

ParkDC: Penn Quarter/Chinatown

DDOT is conducting a pilot study to implement demand-based pricing for on-street parking in the Penn Quarter and Chinatown neighborhoods in downtown Washington, D.C. DDOT is applying an "asset-lite" approach to occupancy detection and prediction that seeks to fuse data from multiple sources, including in-ground parking sensors, fixed and mobile cameras, computer vision, enforcement and citation issuance, and meter payment transactions. Occupancy and hourly meter pricing and policies are available to customers through mobile applications developed by DDOT and by private app developers using an available data feed. The main objective of the program was to improve the parking experience for customers, encourage the effective use of curbside space, and reduce downtown congestion associated with parking activity for personal vehicles, motorcoaches, and freight vehicles. While efforts related to motorcoach operations were shelved, freight management efforts moved forward using the results of the PBC and parking violation data review as a starting point.

The study area, represented in Figure 12, comprises three partially overlapping D.C. neighborhoods: Downtown, Chinatown, and Penn Quarter. The area is in the heart of D.C., with the National Mall to the south, the White House to the west, the Convention Center to the north, and Union Station to the east. Popular destinations in the area include the Capital One Arena, National Portrait Gallery, Chinatown, and a vibrant commercial district, all creating a consistently high demand for parking as well as commercial loading and unloading. From a transportation perspective, the study area includes 114 block faces (91 with parking meters), approximately 900 metered parking spaces, 30 loading zones, six Capital Bikeshare stations, two on-street car-sharing spaces, three Metrorail stations serving all regional Metrorail lines, Metrobus stops, and freeway-arterial interactions to the east of the study area. One unique aspect of the neighborhood is the 500 block of 10th Street, NW (10th Street NW between E Street, NW and F Street, NW). Because of the tourist attractions on this block (Ford's Theater, Petersen House, and souvenir stores), the block does not allow passenger vehicle parking, but does have three loading zones, including a motorcoach zone.

Relationship between Parking Availability and Delivery Vehicles. DDOT sought to improve multimodal conditions for all travelers in the Penn Quarter/Chinatown pilot area, including delivery vehicles. When parking demand outweighs supply, drivers are motivated to park

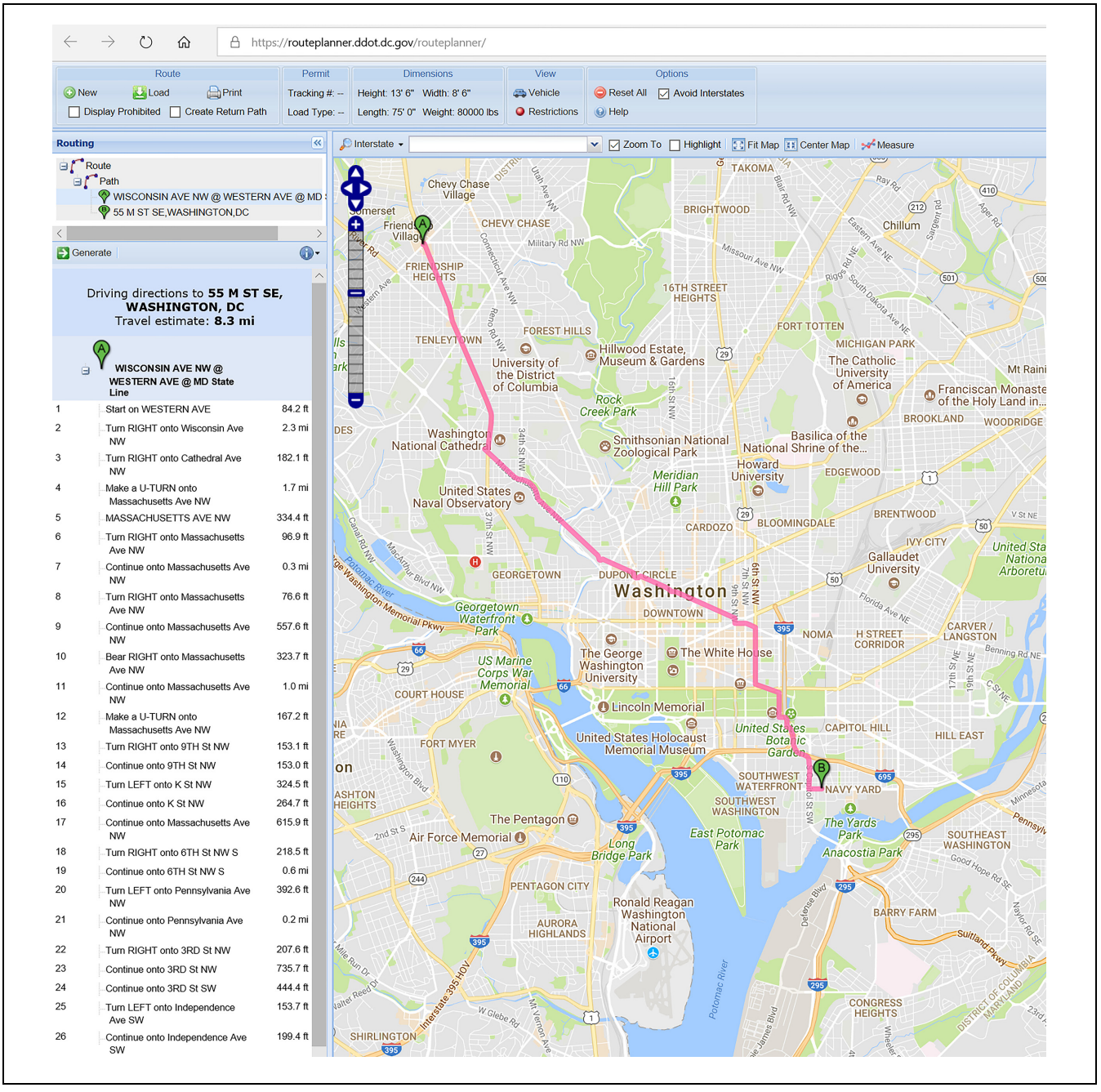


Figure 11. DDOT OSOW vehicle route planner.

illegally or to encroach on loading zones. Thriving businesses in downtown districts must receive deliveries, and delivery truck parking and idling can block travel lanes if loading zones are already occupied. An example is shown in Figure 13.

DDOT conducted video surveys using time-lapse cameras of loading zone activity in 2016 before the first demand-based price change and after the fifth price change was implemented in 2017. The time-lapse cameras were used to gather one week of occupancy data at each loading zone in the study area. Building on the PBC and parking violation data, the project team collected information about the types of vehicles that used each loading zone (passenger, commercial, or bus), and the duration of all vehicle parking sessions. DDOT also collected vehicle type and session duration information for double-parking events at each loading zones. The intent of the data collection was to determine if the

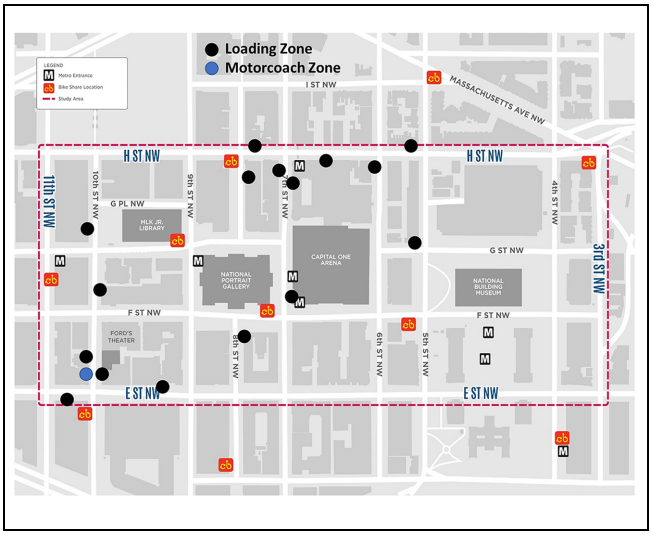


Figure 12. The Penn Quarter/Chinatown study area and loading zones.



Figure 13. Delivery vehicle activity in the study area.

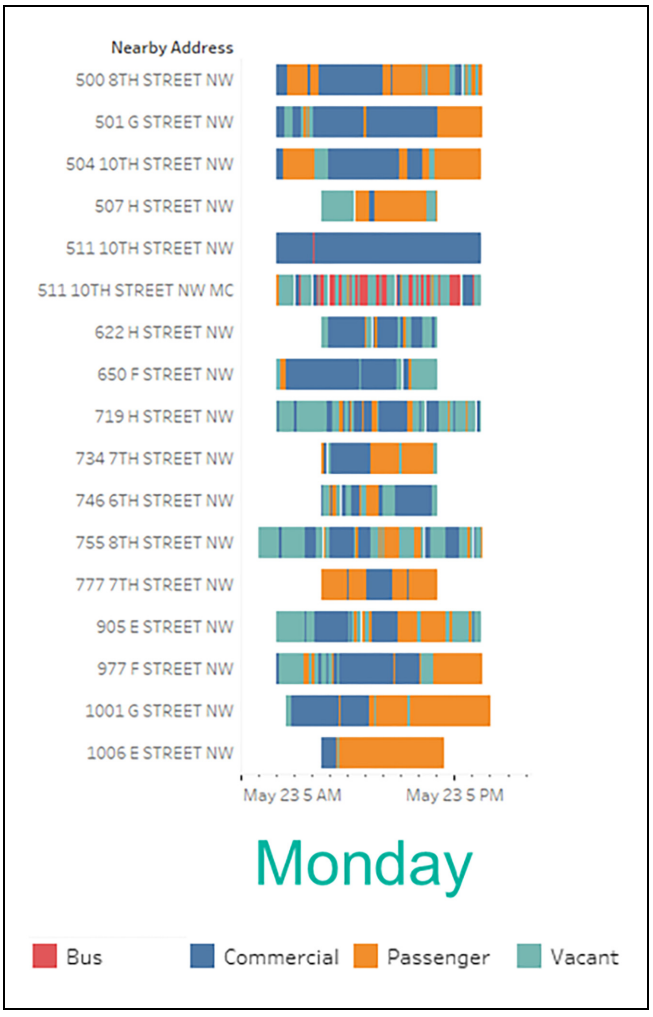


Figure 14. Loading zone activity by vehicle type on a sample day in 2016.

amount of unauthorized vehicle use was consistent with the previous data review efforts (the previous efforts were not able to account for the use of the loading zone by vehicles with one-day or annual permits), and to identify the magnitude of the double-parking issue.

At the time of the 2016 video survey, loading zones were priced at the prevailing rate of $2.30 per hour and were in operation until 7:00 p.m. DDOT compared the vehicle type and duration of all parking events in each loading zone to determine whether certain loading zones were experiencing high volumes of unauthorized use. As shown in Figure 14, many of the loading zones in the pilot area experienced unauthorized passenger vehicle activity throughout the weekday. When multiple vehicle types used the loading zone at the same time, the figure denotes the first vehicle parked, which may hide some overlapping vehicles. Eight of the 16 loading zones observed in the before conditions assessment were occupied by passenger vehicles more than 50% of the time they were in operation. Thirteen of the 16 loading zones experienced a greater number of unique passenger vehicle parking sessions than all other vehicle types.

When looking at individual loading zones in more detail, shown in Figure 15, it becomes apparent that the use of loading zones by passenger vehicles is not confined to any one location. Further, a review of the double-parking data showed that 13 of the 16 loading zones experienced double parking throughout the week with double-parking sessions ranging in length from 5 min to 8 h.

Using previous information from the PBC and parking violation analysis along with more recent data from the parkDC pilot area, DDOT determined that proactive measures were necessary to discourage the unauthorized use of loading zones in the pilot area. DDOT raised the hourly parking rate at all loading zones to match the

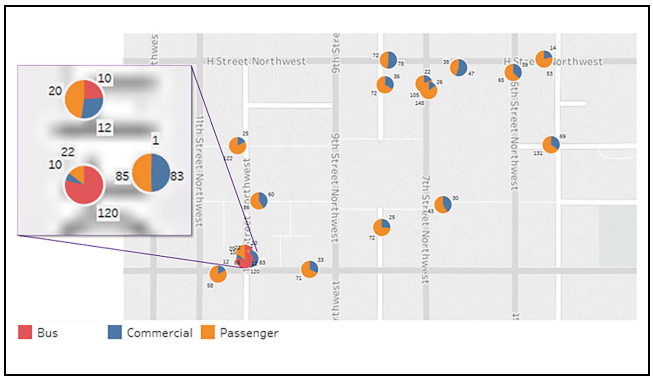


Figure 15. Total vehicles parked in individual loading zones (2016).

highest prevailing on-street parking rate on their associated block faces and loading zone hours of operation were extended from 7:00 to 10:00 p.m. If the highest prevailing on-street rate on a given block face was $4.50 during midday, then the loading zone on that block would be priced at $4.50 per hour throughout the entire day. This higher price was intended to serve as a disincentive to passenger vehicles and other unauthorized users, and the extended hours of operation were intended to improve accessibility for delivery vehicles attempting to access the study area during off-peak hours. While DDOT could have also increased the fine for unauthorized use of the loading zone further in the parkDC: Penn Quarter, Chinatown performance parking zone (because of D.C. Council legislation allowing this action within performance parking zones), DDOT chose not to do so as part of the parkDC pilot because of potential negative public reaction.

Performance Parking in Loading Zones. Additional loading zone data was collected in late 2017 so the pilot team could assess the results of its loading zone pricing strategy. Again, using one week of time-lapse camera footage, DDOT found that while the number of unique instances of double parking increased by 13% after prices increased, the total amount of time vehicles spent double parking decreased by 43% (Table 1). More follow-up data is needed, however, because of the relatively small sample size and several outliers, particularly on the 500 block of 10th Street, NW, which does not allow passenger vehicle parking. DDOT intends to build on the preliminary findings from the parkDC pilot to grow its loading zone pricing and enforcement program, recognizing that a robust program has the potential to shift delivery and other commercial trips to off-peak periods and reduce instances of double parking.

In addition to the time-lapse camera footage, DDOT also reviewed the number of citations given in the study area to unauthorized vehicles parked in loading zones. However, inconsistent enforcement in the study area because of disruptive events (the reconstruction of several Metrorail lines, and large events like the presidential inauguration), along with a different enforcement mechanism in the parkDC pilot area creating issues with enforcement personnel during the pilot, made it

Table 1. Minutes Vehicles Were Observed Double Parked at Loading Zones

	Minutes before (August 2017)	Minutes after (March 2018)	Percent change (%)
504 10th Street, NW	463.7	5.0	–99%
511 10th Street, NW	398.6	100.0	–75%
905 E Street, NW	125.0	80.0	–36%
501 G Street, NW	110.0	115.0	5%
719 H Street, NW[a]	64.9		
511 10th Street, NW (MC)[a]	55.1		
746 6th Street, NW[a]	35.1		
977 F Street, NW	30.0	40.0	33%
1006 E Street, NW	20.1	40.0	99%
500 8th Street, NW[a]	15.0		
755 8th Street, NW	15.0	20.0	33%
777 7th Street, NW	15.0	225.0	1400%
650 F Street, NW	5.0	45.0	800%
1001 G Street, NW[a]		65.0	
507 H Street, NW[a]		5.0	
Total minutes (only values with data in both time periods)	1182.4	670.0	–43%
Average (only values with data in both time periods)	131.4	74.4	

[a]Not included in analysis.

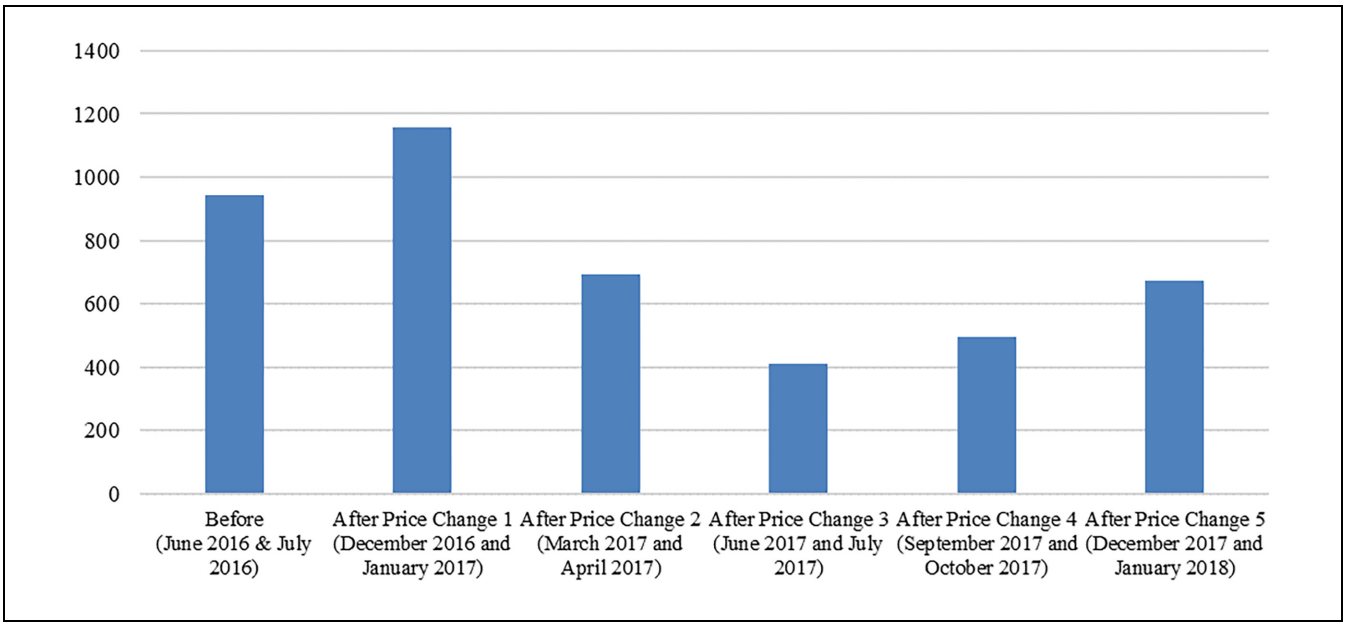


Figure 16. Citations for unauthorized vehicles parked in loading zones.

impossible to draw conclusions from the citation data. Nevertheless, the number of citations for unauthorized vehicles parked in loading zones is provided in Figure 16.

Future Curbside Freight Management Efforts in the District

The ability to collect and analyze data pertaining to the commercial vehicle loading zones and their usage enables the District to continue to improve the commercial vehicle loading zone program and better understand overall curbside usage. DDOT has identified several next steps, including:

- Increased and targeted loading zone enforcement
- Increased disincentives for violations
- Data-driven modifications to the loading zone program
- Modifications to PBC program.

Loading Zone Enforcement

DDOT has often heard complaints from the motor carrier industry regarding personal vehicles in loading zones, which was confirmed by the data presented in this paper. With this data, DDOT will continue to work with enforcement partners in the District to continue targeted enforcement of unauthorized vehicles where they are known to occupy loading zones illegally. Other tools like time-lapse cameras can supplement and act as a calibration for the data collected currently, as they will fill in any gaps in usage the data does not present (if a commercial vehicle has an annual or day pass, for instance).

Disincentivize Violations

With the understanding that unauthorized vehicles make up 86% of all loading zone transactions, the District will evaluate the possibility of raising the fine for unauthorized vehicles. When the District began charging for the use of all curbside commercial vehicle loading zones in January 2015, the District raised the fine for an unauthorized vehicle in a loading zone from $50 to $100. While DDOT could increase the fine further in the parkDC: Penn Quarter, Chinatown performance parking zone (because of D.C. Council legislation allowing this action within performance parking zones), DDOT chose not to do so as part of the parkDC pilot because of potential negative public reaction. The District will evaluate whether an additional increase in the fine would provide a further deterrent to the unauthorized use of loading zones. For the fine increase to be a successful deterrent, however, coordination with enforcement agencies will be necessary.

Data-Driven Program Modification

Data gathered from loading zone usage also can be incorporated into the existing LZAM. The model can be calibrated with comprehensive loading zone usage data. Acquiring this comprehensive data set will require

changes to the loading zone program as it exists, since it does not capture loading zone usage by commercial vehicles with an annual or day pass. The ongoing freight trip generation study will also provide valuable data on loading activities to feed into future versions of the model.

Modifications to PBC Program DDOT will work with enforcement partners to understand the feasibility, cost, and implementation logistics of restricting PBC transactions in loading zones to legitimate vehicles. While this effort likely has the highest potential return on investments, challenges include:

- *Defining what is and is not a commercial vehicle*: The current program requires a U.S. Department of Transportation (USDOT) number to purchase an annual or day pass, although DDOT understands that many small businesses deliver in vans that do not have USDOT numbers. These vehicles, however, can pay at the meter or through the PBC option. Requiring a commercial license plate is another option, though this requires knowledge of how all states plate their commercial vehicles. Alternatively, the District can designate certain loading zones for use by large vehicles with a USDOT number and others for use by smaller commercial vehicles that do not have USDOT numbers. However, this option would require the further designation and delineation of loading zones, a loading zone policy change, and additional enforcement.
- Require annual and daily permit holders to initiate a no-cost transaction via the PBC provider: While this would allow DDOT to collect additional data pertaining to loading zone usage, it would require curbside commercial vehicle loading zone users to perform an extra action.

Conclusion

Demands on the curbside in an urban area like the District are many and varied and will only continue to increase as the city grows economically and in population. Commercial vehicle loading and unloading is a crucial part of urban curbside operations. To better manage the many demands on curbside use, the District has implemented a variety of projects and programs. These include metering or otherwise pricing loading zones and parking, expanding the size and increasing the number of loading zones, increasing the number of metered parking spaces, extending loading and metering time periods, simplifying and standardizing truck parking rules, and encouraging nighttime or off-peak goods delivery (*4*).

- The PBC data, feedback from the carrier industry, and time-lapse camera data collected as part of the parkDC efforts all show that loading zones are often occupied by unauthorized vehicles, requiring large vehicles to double park to load and unload.
- The District will continue to collect and evaluate data currently available, while continuing to improve data collection through programmatic changes enabling a more comprehensive view of loading zone activity. This additional detail will allow for a more detailed evaluation and better management of loading zone activity and curbside space allocation.
- Further coordination with enforcement partners to restrict PBC transactions in loading zones to legitimate vehicles is a DDOT priority, although technological and logistical hurdles remain.

As gaps in understanding curbside movement close with increased data collection and time-lapse videos, DDOT plans to evolve the programs associated with curbside management to reflect discoveries and observations resulting from the curbside data. This complete visibility of curbside loading zone activity, without requiring policy or program determinations mentioned above, allows DDOT to develop smarter policy, program, and management practices. Improved curbside management will help reduce congestion through fewer instances of double parking and curbside conflicts, thereby facilitating increased economic development and maintaining economic vitality.

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Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: SSD, BOP, LR, AP, SD, and MD; data collection: BOP, MS, and ECS; literature review: LR; analysis and interpretation of results: SSD, BOP, AP, MS, MD, and ECS; draft manuscript preparation: AP. All authors reviewed the results and draft manuscript, and approved the final version of the manuscript.

References

1. *District of Columbia Freight Plan*. District Department of Transportation. 2014. http://www.godcgo.com/Portals/0/Freight_PDF/District-of-Columbia-Freight-Plan-Final-Report-10-15-2014.pdf. Accessed July 21, 2015.

2. *Mayor's Parking Taskforce Report*. Mayor's Parking Taskforce Committee Members. 2003. https://comp.ddot.dc.gov/Documents/Mayor's%20Parking%20Task%20Force%20Report.pdf. Accessed July 29, 2015.
3. *Downtown Congestion Task Force Final Report*. Downtown Congestion Taskforce. 2004. http://www.mwcog.org/uploads/committee-documents/vFxeWVY20050128141638.pdf. Accessed July 29, 2015.
4. *District Department of Transportation Curbside Management Study*. District Department of Transportation. 2014. https://comp.ddot.dc.gov/Documents/District%20Department%20of%20Transportation%20Curbside%20Management%20Study.pdf. Accessed July 29, 2015.
5. Alho, A. R., and J. A. Silva. Analysis of the Relation between Land-Use/Urban Freight Operations and the Need for Dedicated Infrastructure/Enforcement – Application to the City of Lisbon. 13th World Conference on Transportation Research, 2013 Rio de Janeiro, Brazil. http://www.wctrs.leeds.ac.uk/wp/wp-content/uploads/abstracts/rio/general/1284.pdf. Accessed July 15, 2015.
6. Robbins, G., and B. M. Yee. San Francisco: Then and Now Downtown Pick-Up and Delivery. *Proc., ITE 2002 Annual Meeting and Exhibit*, Philadelphia, Pa., 2002, p. 18.
7. Jaller, M., J. Holguin-Veras, and S. D. Hodge. Parking in the City. *Transportation Research Record: Journal of the Transportation Research Board*, 2013. 2379: 46–56.
8. Rinnan, M. D. Chicago CBD Loading Zone Policies – Looking Beyond the Curb. *Proc., Institute of Transportation Engineers 2008 Annual Meeting and Exhibit*, Anaheim, Calif., 2008, p. 14.
9. *Schedule of Requirements for Loading Berths, Loading Platforms, and Service/Delivery Loading Spaces*. Rule 11-2201. District of Columbia Municipal Regulations, 2015.
10. Schaller, B., T. Maguire, D. Stein, W. Ng, and M. Blakeley. Parking Pricing and Curbside Management in New York City. Presented at 90th Annual Meeting of the Transportation Research Board, Washington, D.C., 2011.
11. Zalewski, A. J., S. Buckley, and R. Weinberger. Regulating Curb Space: Developing a Framework to Understand and Improve Curbside Management. Presented at 91st Annual Meeting of the Transportation Research Board, Washington, D.C., 2012.
12. Dablanc, L., G. Giuliano, K. Holliday, and T. O'Brien. Best Practices in Urban Freight Management. *Transportation Research Record: Journal of the Transportation Research Board*, 2013. 2379: 29–38.
13. Nordtomme, M. E., J. Andersen, A. B. Sund, I. Roche-Cerasi, T. Levin, and O. Eidhammer. Green Urban Distribution: Evaluation of Adapted Measures for the City of Oslo. *European Transport Conference* 2013. http://abstracts.aetransport.org/paper/index/id/49/confid/1. Accessed July 15, 2015.
14. Salama, P., A. Lubinsky, K. Kraft, P. Lipson, L. Torres, B. Roy, and P. Windover. *New York City Green Loading Zones Study*. NYSERDA Report C-13-52. New York State Energy Research and Development Authority, New York State Department of Transportation, Albany, 2014.
15. *UPS Launches eBike Solution in Pittsburgh*. United Parcel Service. 2017. https://pressroom.ups.com/pressroom/ContentDetailsViewer.page?ConceptType=PressReleases&id=1510149665708-495. Accessed July 31, 2018.
16. *Truck and Bus through Routes and Restrictions*. District Department of Transportation. 2014. http://www.godcgo.com/truckandbusmap. Accessed July 29, 2015.
17. *LicensePlate.cc*. Bannister Design LLC. Accessed July 21, 2015.
18. District Department of Transportation. Route Planner 2017. https://routeplanner.ddot.dc.gov/routeplanner/

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